

Microeconomic Theory III

ECON 709

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The syllabus...

About Economic Theory (Rubinstein, Introduction)

- ▶ investigation of concepts that we use to think about real-life economic situations with formal/mathematical definitions and models
- ▶ attempt to better understand reality
- ▶ “economic” model is mathematical model with interpretation
- ▶ formality (the math) provides a way to think systematically about economic interactions
- ▶ models are simplistic and “unrealistic”; focus on essential and important elements, not “irrelevant” details; attempt to clearly discern the workings of the essential elements
- ▶ economic theory does not provide you with the “truth”. No prediction of the “future” (always be very skeptical about any policy recommendations!). Models are always an abstraction aimed at elucidating some economic situation.
- ▶ Every model is based on the assumptions of its creator and is subjective. It is not necessarily the “right” model.

The economic theory we see in this course...

Part A: Choice

- ▶ discussion of formal concepts and models about the behavior of a single economic agent

Part B: General equilibrium

- ▶ discussion of formal concepts and models about the sum of behaviors of single economic agents in “large” markets (no strategic interaction between agents)

Preferences

based on: Rubinstein, lecture 1; MWG, section 1.B

Preferences

- ▶ in order to study (rational) choice or "satisfaction", let's introduce a formal concept of tastes/opinions/attitudes/preferences...
- ▶ let X be a set of alternatives
- ▶ economists summarize taste/opinion/attitude/preferences etc. over X in a *preference relation*
- ▶ a *preference relation* on X is denoted by $\succsim$

A preference relation is a binary relation

- ▶ a preference relation $\succsim$ on X is a *binary relation* on X which allows us to **compare pairs of alternatives** $x, y \in X$,
 - ▶ $x \succsim y$ means "x is at least as good as/weakly preferred to y"
- ▶ the binary relation $\succsim$ is defined for every pair $x, y \in X$
- ▶ this means that for every pair $x, y \in X$, the question "Is x at least as good as y?" ($x \succsim y$?) is answered with either "Yes" ($x \succsim y$) or "No" (not $x \succsim y$; $x \not\succsim y$).
- ▶ thus, given a preference relation, for any pair of alternatives $x, y \in X$ (not necessarily distinct), exactly one of four possibilities holds:
 - ▶ $x \succ y$ and $y \not\succsim x$
 - ▶ $x \succsim y$ but not $y \succ x$
 - ▶ $y \succ x$ but not $x \succ y$
 - ▶ neither $x \succ y$ nor $y \succ x$

Strict preference and indifference relation

- ▶ two other binary relations are associated with a *weak* preference relation $\succsim$:
the *strict preference relation*, denoted by $\succ$,
and the *indifference relation*, denoted by $\sim$
- ▶ “ $x \succ y$ ” means “ x is strictly better than y ”
- ▶ “ $x \sim y$ ” means “ x is exactly as good as (indifferent to) y ”
- ▶ given a weak preference relation $\succsim$ on X , the associated
 - ▶ *strict preference relation*, $\succ$, on X is defined by:
$$x \succ y \Leftrightarrow x \succsim y \text{ and not } y \succsim x$$
 - ▶ *indifference relation*, $\sim$, on X is defined by:
$$x \sim y \Leftrightarrow x \succsim y \text{ and } y \succsim x$$

“Rational” preference relations

In economics, we usually assume that a preference relation is “rational”, this means that the preference relation between any two alternatives is well-defined and satisfies an additional “consistency” criterion.

Definition 1

A preference relation $\succsim$ on a set X is *rational* if it satisfies the following three properties:

1. *Reflexivity*: For any $x \in X$, $x \succsim x$.
2. *Completeness*: For any distinct $x, y \in X$, $x \succsim y$ or $y \succsim x$.
(“or” does not exclude that both hold)
3. *Transitivity*: For any $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.

Before the discussion of these properties, let's see what they imply for the strict and the indifference relation...

Implications of Rationality of $\succsim$ for $\succ$ and $\sim$

Proposition 1

If $\succsim$ is rational then:

1. $\succ$ is irreflexive ($x \succ x$ never holds) and transitive (if $x \succ y$ and $y \succ z$, then $x \succ z$).
2. $\sim$ is reflexive ($x \sim x$ for all $x \in X$), transitive (if $x \sim y$ and $y \sim z$, then $x \sim z$) and symmetric (if $x \sim y$, then $y \sim x$)
3. if $x \succ y \succsim z$, then $x \succ z$.

Proof of Proposition 1, 3.

3. If $x \succ y$, then $x \succsim y$ by def. of $\succ$. Then by transitivity of $\succsim$, $x \succsim z$. Suppose to the contrary that also $z \succsim x$ (implying $x \sim z$ and not $x \succ z$). Since $y \succsim z$, transitivity of $\succsim$ then implies that $y \succsim x$ (and hence $x \sim y$), a contradiction to $x \succ y$. Thus $x \succ z$.



Proof of Proposition 1, 1. and 2.

1. "If $\succsim$ is rational, then $\succ$ is irreflexive ($x \succ x$ never holds) and transitive (if $x \succ y$ and $y \succ z$, then $x \succ z$)."

By reflexivity of $\succsim$, $x \succsim x$ for all $x \in X$, and thus $x \succ x$ never holds. If $x \succ y$ and $y \succ z$, then $x \succ y \succsim z$. Thus $x \succ z$ by Prop. 1.3.

2. "If $\succsim$ is rational, then $\sim$ is reflexive ($x \sim x$ for all $x \in X$), transitive (if $x \sim y$ and $y \sim z$, then $x \sim z$) and symmetric (if $x \sim y$, then $y \sim x$)."

By reflexivity of $\succsim$, $x \succsim x$ for all $x \in X$, and thus $x \sim x$ always holds. If $x \sim y$ and $y \sim z$, then $x \succsim y$, $y \succsim x$, $y \succsim z$, and $z \succsim y$. Thus $x \succsim z$ and $z \succsim x$ by transitivity of $\succsim$, and hence $x \sim z$. If $x \sim y$, then $x \succsim y$ and $y \succsim x$. Thus $y \succsim x$ and $x \succsim y$, and $y \sim x$.



Are the “rationality” properties of $\succsim$ sensible assumptions?

Reflexivity: For any $x \in X$, $x \succsim x$.

- ▶ means that when comparing an alternative with itself, the alternative is at least as good as itself
- ▶ implies indifference between two identical options, i.e. $x \sim x$

Completeness: For any distinct $x, y \in X$, $x \succsim y$ or $y \succsim x$.

- ▶ completeness implies that the preference relation is well-defined for any two distinct alternatives; that is, for any pair of distinct x and y in X , we have one of the three mutually exclusive possibilities:
 - ▶ $x \succsim y$ and not $y \succsim x \Leftrightarrow x \succ y$ (x is strictly better than y)
 - ▶ $y \succsim x$ and not $x \succsim y \Leftrightarrow y \succ x$ (y is strictly better than x)
 - ▶ $x \succsim y$ and $y \succsim x \Leftrightarrow x \sim y$ (x is as good as y)
- ▶ any two distinct alternatives x and y can be ranked
- ▶ this is a strong assumption:
 - ▶ requires you to always know your preference
 - ▶ even though alternatives might not be well known to you
- ▶ but useful: how could choice/satisfaction over alternatives be analyzed if preferences were undefined?

Transitivity: For any $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.

Examples:

- ▶ If "large" is preferred over "medium", and "medium" over "small", then "large" is also preferred over "small". This seems plausible.
- ▶ If "red" is preferred over "blue" and "blue" over "yellow", then "red" should also be preferred over "yellow". This is less obvious.

Transitivity is a strong assumption. We can easily think of plausible intransitive preference relations...

Potential Sources of Intransitivity

1. Deriving a preference relation over alternatives by aggregating preference relations over components:

- ▶ suppose alternatives in X consist of several components and an individual derives her preference relation $\succsim$ over X by aggregating her preferences relations for the different components, for example
" $x \succsim y$ iff a majority of the components in x is preferred over the components in y "

- ▶ e.g.:

- ▶ set of holiday packages $X = \{a, b, c\}$ and each $i \in X$ is characterized by its price p_i , location l_i , and food f_i
- ▶ assume the following preference relations over prices, locations, and food:

$$p_a \succ_p p_b \succ_p p_c$$

$$l_b \succ_l l_c \succ_l l_a$$

$$f_c \succ_f f_a \succ_f f_b$$

- ▶ if above rule is applied, then $a \succ b$, $b \succ c$, and $c \succ a$, and thus the preference relation over X violates transitivity

Potential Sources of Intransitivity

2. Use of similarities to establish indifference:

- ▶ suppose an individual expresses indifference between two alternatives that are too "close" to be distinguishable for her
- ▶ example:
 - ▶ let X be the set of real numbers
 - ▶ consider an individual whose attitude toward the alternatives is "the larger the better" however she cannot determine whether $x > y$ unless the difference is at least 1
 - ▶ thus $x \succ y$ if $x \geq y + 1$ and $x \sim y$ if $|x - y| < 1$
 - ▶ then $1.5 \sim .8$ and $.8 \sim .3$ but $1.5 \succ .3$ so it is not true that $1.5 \sim .3$, violating transitivity

Then why do we assume transitivity?

- ▶ We avoid "cycles" in preferences, like $a \succ b$, $b \succ c$, and $c \succ a$.
- ▶ With cycles, there exists a sequence of pairwise choices such that the decision maker always prefers the "newly" offered alternative.
- ▶ With cycles, we might not be able to determine the set of "best" alternatives from X . Determining a "best" alternative is important for choice analysis.
- ▶ And in many situations, transitivity is a natural property of preferences...

Implicit Assumptions on Preference Relations

- ▶ no intensities of preferences:
a preference relation cannot capture the difference between "I somewhat prefer x over y " vs. "I love x and hate y "
- ▶ no dependency on other factors:
e.g. it is impossible to, depending on circumstances, sometimes prefer x over y , and sometimes y over x

Equivalent Definitions of Rational Preference Relations

1. combining reflexivity and completeness, e.g. MWG:
 - ▶ "A preference relation is rational if it satisfies 1) Transitivity, and 2) Completeness: 'For any $x, y \in X$, $x \succsim y$ or $y \succsim x$ '."
 - ▶ The definition of completeness is extended and includes reflexivity. Different ways of saying one thing!
2. avoiding the terminology "rational" (e.g. Rubinstein):
 - ▶ "A binary relation $\succsim$ is a preference relation if $\succsim$ is 1) reflexive (R), 2) complete (C), and 3) transitive (T)."
 - ▶ according to this definition, every preference relation is "rational" (R, C, T), and if a binary relation $\succsim$ is not "rational", then it is not a preference relation
 - ▶ this avoids classifying R, C, and T as "rational" properties
 - ▶ terminology "rational" is debated because you can think of "sensible" tastes/opinions not satisfying R, C, T
 - ▶ it gives up calling a binary relation $\succsim$ a preference relation if $\succsim$ is not R, C, T
 - ▶ in the end, only a different terminology is used here

We'll stick to Definition 1...